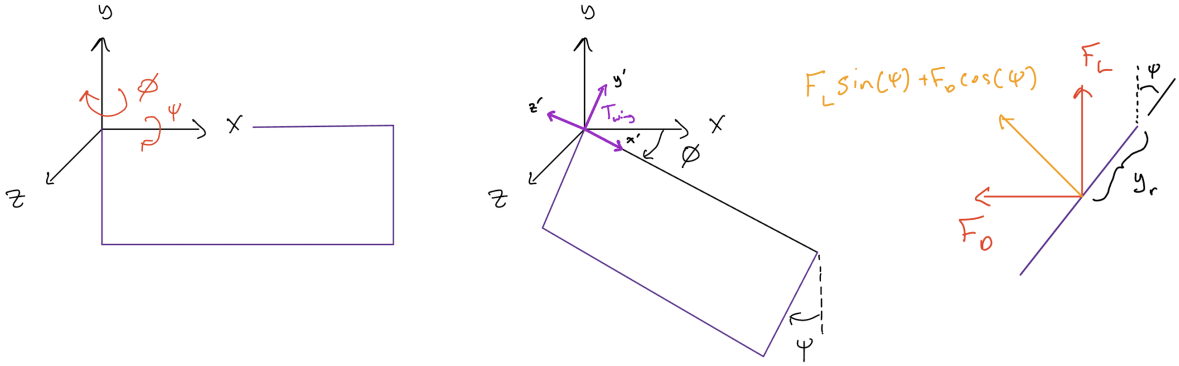


MAE 269: HW 5

Due Wednesday May 20, 2025 at midnight

1 Modeling passive pitching flapping wings

Many small robots and some insects have wing hinges that passively rotate. In this HW we will simulate the motion of a passively pitching wing with modeled aerodynamic forces acting on it.



We will model the flapping wing as a thin plate (with ≈ 0 thickness in z direction). The wing rotates around the y axis as the stroke plane with stroke angle ϕ (which we control). The wing passively pitches to an angle ψ about the x -axis. Thus, the transformation that takes us from world frame to a frame attached at the wing root is

$$T_{wing} = \begin{bmatrix} R_y(\phi)R_x(\psi) & 0 \\ 0 & 1 \end{bmatrix} \quad (1)$$

Because we are not locating the wing reference frame at the center of mass, we have to consider off-diagonal inertia terms and so we can use the generalized inertia tensor

$$\mathcal{G}_{wing} = \begin{bmatrix} I_{xx} & -I_{xy} & 0 & 0 & 0 & 0 \\ -I_{xy} & I_{yy} & 0 & 0 & 0 & 0 \\ 0 & 0 & I_{xx} + I_{yy} & 0 & 0 & 0 \\ 0 & 0 & 0 & m & 0 & 0 \\ 0 & 0 & 0 & 0 & m & 0 \\ 0 & 0 & 0 & 0 & 0 & m \end{bmatrix} \quad (2)$$

where all the z -terms have been set to zero because of the small thickness and all off diagonal terms are negative.

1. Use the standard Euler-Lagrange method to derive the equations of motion for the ψ variable. Verify that the equation of motion is the same as equation 2.16 of the first reference of the optional reading

$$I_{xx}\ddot{\psi} = M_x + I_{xy}\ddot{\phi}\cos\psi + \frac{1}{2}I_{xx}\dot{\phi}^2\sin 2\psi$$

where the moment about the rotational hinge, M_x , has two contributions:

$$M_x = M_{elastic} + M_{aero}$$

2 The nonlinear oscillations of the insect flight motor

Many insects that have fast wingbeats (bees, flies, and mosquitoes for example) use a method of muscle actuation called “Asynchronous”. An asynchronous muscle does not actuate by a signal from the brain, but instead has a special mechanical property that when it is stretched, it generates a delayed restoring tension. This process is called delayed stretch activation (dSA). These asynchronous muscles pull against the insect exoskeleton which oscillates the wings through an elastic hinge, and thus the dSA coupled to spring-wing mechanics leads to self-excited wingbeats: the hallmark of a limit-cycle.

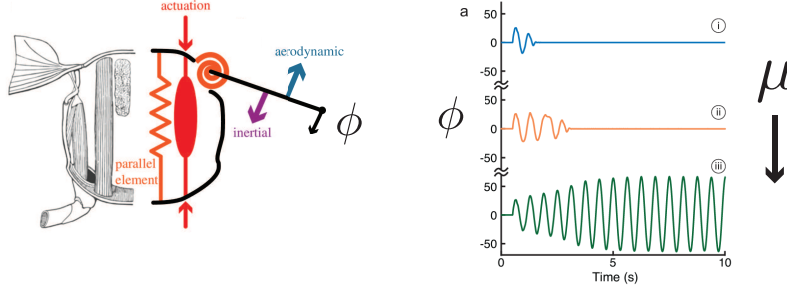


Figure 1: Left: cross section of an insect thorax, right: onset of limit-cycle oscillations as parameter μ is increased.

A simplified model of delayed stretch activation coupled to spring-wing dynamics is given by the following equations:

$$I\ddot{\phi} + \Gamma|\dot{\phi}|\dot{\phi} + \frac{k\phi}{T^2} = \frac{F}{T} \quad (3)$$

$$\ddot{F} + r_3(1 + \kappa)\dot{F} + \kappa r_3^2 F = -\mu\kappa r_3^2 \dot{\phi} \quad (4)$$

The first equation defines the angular movement of the wing (denoted by wing angle ϕ), with parameters including wing inertia I , aerodynamic coefficient Γ , spring constant k , and gear ratio T . The second equation determines the dynamics of muscle actuation (defined by F) with parameters of muscle rate r_3 , actuation shape factor κ ($0 < \kappa < 1$), and muscle strength μ .

Assume that κ , I , Γ , k , and T are fixed parameters and the muscle properties r_3 and μ can vary. The state vector of the system is $\vec{x} = [\phi, \dot{\phi}, F, \dot{F}]^T$.

1. Turn the system of equations into four first order differential equations $\frac{d\vec{x}}{dt} = g(\vec{x})$.
2. Determine the fixed point(s) of this system, $g(\vec{x}^*) = 0$
3. Linearize the system about the fixed point, $\frac{d\vec{x}}{dt} \approx \frac{d}{d\vec{x}}g(\vec{x})|_{\vec{x}=\vec{x}^*}(\vec{x} - \vec{x}^*) \rightarrow A\vec{x}$.
4. Construct the characteristic equation for the matrix A . The characteristic equation results from expanding the equation, $\det(A - \lambda I) = 0$, into a fourth order polynomial in λ with four possible eigenvalue solutions $\lambda_{i=1,\dots,4}$. (Note I is the identity matrix).
5. Solutions to the fourth order characteristic polynomial yield solutions to the eigenvalues. Regimes where an eigenvalue has a negative real component (is stable), to a positive real component (is unstable) are separated by the boundary where λ is purely imaginary ($\lambda = i\omega$). Substitute $\lambda = i\omega$ into the characteristic equation and separate the real and imaginary components into two separate equations.
6. In the $\mu > 0$ vs $r_3 > 0$ parameter space, plot the two equations that are generated from the assumption of $\lambda = i\omega$. These lines separate the parameter space into regimes of limit-cycles and regimes where no oscillations occur.
7. Color code the parameter space plot with red where the real components of at least one eigenvalue is positive, and blue in regions where all eigenvalues have negative real components.

3 Optional paper review

Please read through the required reading. Provide your comments on the required paper. Describe anything that you didn't understand in the paper.

- Whitney, J. P., & Wood, R. J. (2010). Aeromechanics of passive rotation in flapping flight. *Journal of Fluid Mechanics*, 660, 197–220.
- Buchli, J., Righetti, L., & Ijspeert, A. J. (2006). Engineering entrainment and adaptation in limit cycle systems : From biological inspiration to applications in robotics. *Biological Cybernetics*, 95(6), 645–664.
- Lynch, J., Gau, J., Sponberg, S., & Gravish, N. (2022). Autonomous Actuation of Flapping Wing Robots Inspired by Asynchronous Insect Muscle. *IEEE International Conference on Robotics and Automation (ICRA)*.
- Lynch, J., Gau, J., Sponberg, S., & Gravish, N. (2021). Dimensional analysis of spring-wing systems reveals performance metrics for resonant flapping-wing flight. *Journal of the Royal Society, Interface / the Royal Society*, 18(175), 20200888.